MILITARY SPECIFICATION

CHARGERS, BATTERY, NICKEL-CADMIUM, AIRCRAFT, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements of airborne battery chargers for use with aircraft nickel-cadmium batteries to maintain the batteries in an optimum state of charge condition (see 6.1).

1.2 Classification. The battery chargers will include the following types:

Type I - 28 VDC Input Power
Type II - 115/200 VAC 400 Hz 3-Phase Input Power

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416 Plating, Cadmium (Electrodeposited)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Engineering Specifications and Standards Department (ESSD), Code 93, Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6140

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SPECIFICATIONS (Continued)

Military

MIL-P-116	Preservation - Packaging, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-C-6021	Castings, Classification and Inspection of
MIL-R-6106	Relay, Electric, Aerospace
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-15024	Plate, Tags and Bands for Identification of Equipment
MIL-A-21180	Aluminum Alloy Castings, High Strength
MIL-I-45208	Inspection System Requirements
MIL-E-81910	Electric Power and Control Equipment, Aircraft, General Specification for

(See Supplement 1 for applicable specification sheets)

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking for U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for Selection

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STANDARDS (Continued)

Military (Continued)

MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-45662	Calibration Systems Requirements
MS33543	Criteria-Temperature and Altitude Range, Self-Cooled Electric Equipment

HANDBOOKS

Military

MIL-HDBK-217	Reliability Prediction of Electronic Equipment

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Battery chargers furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.4 and 6.3).

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3.3 Materials, components and processes.

3.3.1 Selection of parts and materials. Specifications and standards for all materials, parts, Government certification and approval of processes and equipment, which are not specifically designated herein shall be selected in accordance with MIL-STD-143. The charger shall not use any gas or vacuum devices, or devices which depend on elevated temperature for operation.

3.3.1.1 Threaded parts. All threads shall conform to MIL-S-7742. All internal or external parts which are threaded shall be positively locked. Accidental loosening of threaded elements shall be prevented by self-locking nuts, safety wiring, or other approved methods.

3.3.1.2 Standard parts. Military standard parts shall be used whenever they are suitable for the purpose, and shall be identified on the drawing by their appropriate military part number. Any part of the same part number shall be interchangeable as applied within the system. Standard parts shall be in accordance with MIL-E-5400.

3.3.1.3 Approval of non-standard parts. Request for approval for the use of non-standard parts shall be in accordance with MIL-E-5400.

3.3.1.4 Commercial (hardware) parts. Commercial utility hardware items, such as screws, bolts, nuts and cotter pins may be used without approval provided they possess suitable properties and are replaceable by MS or AN parts without alteration, and provided the corresponding MS or AN part numbers are referenced in the parts list and on the contractor's drawings.

3.3.2 Material properties.

3.3.2.1 Fungus. Materials used shall be fungus inert.

3.3.2.2 Toxic. Materials which produce harmful toxic effects under any conditions, including fire, shall not be incorporated in the design of the charger.

3.3.2.3 Flammable. Materials which will support combustion shall not be used. Definitions and restrictions on the use of flammable materials shall be in accordance with MIL-STD-454, Requirement 3.

3.3.3 Metals.

3.3.3.1 Corrosion. Metals shall be of a corrosion resisting type or suitably processed to resist corrosion.

3.3.3.2 Magnesium. Magnesium shall not be used.

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3.3.3.3 Dissimilar metals. The use of dissimilar metals in contact with each other (particularly brass, copper, or steel in contact with aluminum or other alloys) shall be avoided. Where contact between dissimilar metals is unavoidable, they shall be suitably protected against electrolytic corrosion. When protection is used, it shall be a type that offers a low impedance path to radio frequency current. (Dissimilar metals are defined in MIL-STD-889.)

3.3.3.4 Castings. Castings shall be in accordance with MIL-C-6021 and shall be Class 2, grade C or better. Aluminum alloy castings shall be in accordance with MIL-A-21180. Defects not materially affecting the suitability of the castings may be repaired at the foundry or during machining by peening, impregnating, welding or other methods acceptable to the procuring activity. Inspection of repaired castings shall be governed by quality control techniques subject to the approval of the procuring activity.

3.3.3.5 Finishes and protective coatings.

3.3.3.5.1 Aluminum alloy parts. Aluminum alloy parts shall be anodically treated in accordance with MIL-A-8625 or chemically treated in accordance with MIL-C-5541. To provide adequate bonding, the aluminum oxide film formed by this treatment shall be removed from the actual contact area of all surfaces that act as a path for electrical current and from the areas under screws, nuts, etc., used for assembly or mounting purposes.

3.3.3.5.2 Steel parts. Steel parts shall be cadmium plated in accordance with Type II or III of QQ-P-416.

3.3.3.5.3 Color. The charger shall be finished in 37038 black in accordance with FED-STD-595. The finish shall be unaffected by testing required by the detail specification.

3.3.4 Insulating materials. Electrical insulating materials shall be in accordance with MIL-STD-454, Requirement 11. Insulation for power circuits shall be arc-resistant in accordance with MIL-STD-454, Requirement 26.

3.3.5 Encapsulation and potting. Materials for encapsulation and embedment shall be selected and used in accordance with MIL-STD-454, Requirement 47.

3.3.6 Semiconductor devices. Semiconductor devices shall be selected and applied in accordance with MIL-STD-454, Requirement 30. Microelectronic devices shall be selected and applied in accordance with MIL-STD-454, Requirement 64. Semiconductor devices shall conform to MIL-S-19500 and shall be class JANTXV, except that when there is no JANTXV, JANTX parts shall be used. Use of non-standard parts shall be substantiated by reliability data or testing to assure adequate device reliability for the application.

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3.3.7 Capacitors. Capacitors shall be chosen and applied in accordance with Requirement 2 of MIL-STD-454.

3.3.8 Soldering. Soldering shall be in accordance with MIL-STD-454, Requirement 5.

3.3.9 Relays. Relays shall comply with MIL-R-6106.

3.4 Design and construction.

3.4.1 Cooling. The battery charger developed under this specification shall be convection-cooled.

3.4.2 Dimensions and weight. The dimensions and weight shall be as specified on the applicable specification sheet. Where only maximum or minimum dimensions are shown, the part need not have the shape shown, but the part, including all protrusions, shall be contained within the outline shown.

3.4.3 Mounting. Mounting provisions shall be as specified on the applicable specification sheet.

3.4.4 Electric connections.

3.4.4.1 Electrical connectors. Connectors shall be approved military types. The external power and control connections shall be as specified on the applicable specification sheet. Equipment structure shall not be relied on as a current path except for radio noise shielding.

3.4.4.2 Internal electric connections. Internal connections shall be made in accordance with MIL-STD-454, Requirements 1, 5, 10, 17, 19, 20 and 69.

3.4.5 Maintainability. The battery charger shall be so designed as to facilitate assembly, disassembly, location of trouble sources, and maintenance without the aid of special tools.

3.4.6 Interchangeability. Provisions for interchangeability are required to the maximum extent possible and shall be in accordance with Requirement 7 of MIL-STD-454.

3.4.7 Adjustment. The charger shall have no external adjustments.

3.5 Performance.

3.5.1 Electrical performance. The charger shall provide the charge and standby performance specified herein. No single charger failure shall damage the battery or place a continuous load across the battery.

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3.5.1.1 Electric input power characteristics. The charger shall meet the requirements for utilization equipment as specified in MIL-STD-704, except as otherwise noted herein.

3.5.1.2 Power requirements. Input current shall be in accordance with the applicable specification sheet.

3.5.1.3 Turn-on/turn-off transients. Charging shall occur within 7.0 seconds of input power application to the charger. Turn-on and turn-off of the charge mode shall not result in uncontrolled on-off cycling, oscillation, or other evidence of instability when operated at the input power specified herein.

3.5.1.4 Charge performance. The charger shall be capable of charging the battery specified on the applicable specification sheet under all operating and environmental conditions listed herein. Charging shall be terminated automatically when the battery is fully charged.

3.5.1.4.1 Temperature sensor. The charger shall utilize a temperature sensor mounted inside the battery, having characteristics specified on the applicable specification sheet, to regulate the charge termination voltage and to terminate charge with a battery temperature of 62.0 $\pm$ 2.8°C. In the event a high battery temperature occurs, the charger shall provide a turn-off/turn-on temperature differential of 2.8° - 5.6°C. Battery charging shall not occur with a battery temperature below -32.0 $\pm$ 2.8°C.

3.5.1.4.1.1 In the event a short or open circuit exists within the temperature sensing circuit, the charger shall automatically turn off.

3.5.1.4.2 Thermal switch. The charger shall accept an input from one side of a normally open thermal switch with the other side connected to ground. The thermal switch is mounted inside the battery and shall terminate charge upon switch closure. Deactivation of the thermal switch shall cause the charger to reinitiate charge.

3.5.1.4.3 Center tap voltage. The charger shall accept a battery tap voltage input and reduce charge current to less than 1.0 A if the voltage of each battery section differs from normal by more than 1.0 $\pm$ 0.2 volt in a constant-current charge mode, or 0.6 $\pm$ 0.1 volt in a constant-potential charge mode. Normal for this condition is defined as the ratio of the number of cells in each section to the total number of cells in the battery times the battery voltage. The charger shall reduce the charge current to less than 1.0 A within 60.0 seconds of sensing the voltage imbalance condition. The voltage tap lead will include a 2000-ohm resistor, and be connected to the positive terminal of the tenth cell referenced from battery negative. The charger shall not be damaged by one or more shorted cells in the battery.

3.5.1.4.4 Charge current. The charge current shall not exceed 1.5C (C is equal to the 1-hour military capacity discharge rate of the battery in amperes) until the battery potential reaches 15.0 V. Then

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the average charge current during any 1 second period shall not exceed that specified on the applicable specification sheet, regardless of input voltage. At no time shall the charger cause electrolyte spewage within a battery that has properly adjusted electrolyte levels.

3.5.1.4.5 Charging time. The maximum time required for the battery charger to charge the specified battery, when all conditions of charge initiation are met, from 0 volt per cell to 80 percent and 100 percent state-of-charge (SOC) shall be as shown in Table I. This requirement applies to batteries that contain actual capacities of up to 125 percent of rated capacities.

TABLE I. Maximum charge time in hours.

Temperature	80% SOC	100% SOC
49°C	1.5	3.0
25°C	1.0	2.0
-18°C	2.0	4.0
-29°C	2.0	4.0

3.5.1.4.6 Capacity. When charged under the temperature conditions of Table I, and discharged under the temperature and discharge current conditions of Table II, the battery shall be capable of maintaining a cell voltage not less than the minimum specified voltage.

TABLE II. Capacity.

Temperature	Discharge Rate (amperes)	Discharge Time (minutes)		End of Discharge (volts/cell)
		80% SOC	100% SOC	
49°C	1C	40	50	0.95
25°C	1C	48	60	0.95
-18°C	1C	36	45	0.95
-29°C	1C	28	35	0.95

3.5.1.5 Charge inhibit circuit. The charger shall be capable of accepting an input from an external inhibit signal, as defined on the applicable specification sheet. The charger shall terminate charge and close the battery relay control contacts (see 3.5.1.6) upon application of the inhibit signal.

3.5.1.6 Battery relay control. Type I chargers shall provide a pair of normally closed relay contacts (in de-energized state) that are capable of conducting 1 ampere for control of the battery relay.

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a. The relay contacts shall be open whenever the input voltage is above 26.0 ± .3 volts. The two relay contacts shall close within 5-15 milliseconds and charging shall be inhibited whenever the input voltage falls below 21.0 ± 1.0 volts. Once the relay contacts close, they shall remain closed for a time of 4.5 ± 2.5 seconds, regardless of the input voltage. Upon completion of the time delay, the relay contacts shall remain closed if the input voltage is below 26.0 ± 0.3 volts, or be open if the input voltage is above 26.0 ± 0.3 V.

b. The relay contacts shall close whenever an external inhibit signal, as defined on the applicable specification sheet, is applied to the charger regardless of the input voltage. The relay contacts shall open when the inhibit signal is removed from the charger and the minimum input voltage and time delay requirements of 3.5.1.6a are satisfied.

c. For any conditions other than those specified in 3.5.1.6a and b, the battery relay control contacts shall remain open.

3.5.1.7 Fault signal. The charger shall terminate charge, and utilize an internal relay with normally closed contacts (in de-energized state), to provide a signal path of less than 5 ohms, capable of conducting 0.4 ampere, as specified on the applicable specification sheet whenever one or more of the following conditions occur.

a. Overheated battery (see 3.5.1.4.1 and 3.5.1.4.2).

b. Battery temperature less than $-32.0^{\circ} \pm 2.8^{\circ}C$ (see 3.5.1.4.1).

c. Defective temperature sensor (see 3.5.1.4.1.1).

d. Battery cell imbalance (see 3.5.1.4.3).

e. Charger input voltage is less than 21.0 ± 1.0 V.

f. Charger input voltage exceeds maximum operating limits.

g. An external charge inhibit signal has been applied to the charger (see 3.5.1.5).

h. Charger power output connection is open.

3.5.1.8 Efficiency. The overall efficiency of the charger shall be as specified on the applicable specification sheet, when tested in accordance with 4.7.2.3.

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3.5.1.9 Short circuit/open circuit capability. The battery charger shall be capable of withstanding a continuous short circuit or open circuit on the output without damage.

3.5.2 Dielectric strength. The charger shall be so designed that insulation breakdown shall not occur between any terminal (except a terminal that is connected directly to the frame during normal operation) and the frame when tested in accordance with 4.7.2.6. Capacitors and solid state components which would be damaged by this test may be disconnected.

3.5.3 Duty cycling. The charger and battery specified on the applicable specification sheet shall successfully complete the duty cycle test of 4.7.2.7 at ambient conditions of 4.7.1.1.

3.5.4 Operating position. The charger shall be capable of operating in any position.

3.5.5 Environmental. The charger, when subjected to the environmental conditions listed in Table III, shall show no:

a. Dimensional distortion beyond specified limits, or cracking of cases, covers or mounting brackets.

b. Breakdown of insulation, loosening or stripping of metal plating or protective coatings from any component part, or corrosion of metal parts.

c. Mechanical failure of any part.

d. Fungus growth.

e. Deterioration of charger identification markings.

f. Degradation of electrical performance beyond specified limits of 3.5.

TABLE III. Environmental conditions.

Environmental Condition	Test Paragraph
Electromagnetic Compatibility	4.7.2.10
Acceleration	4.7.2.11
Dust	4.7.2.12
Salt Fog	4.7.2.13
Temperature-Altitude	4.7.2.14
Mechanical Shock	4.7.2.15
Fungus	4.7.2.16
Humidity	4.7.2.17
Vibration	4.7.2.18
Explosive Atmosphere	4.7.2.19

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3.5.6 Burn-in. Each charger shall be subjected to a burn-in test as described in 4.7.2.20.

3.6 Reliability. A reliability prediction report of the qualification test hardware shall be prepared for approval of the qualifying activity 60 days prior to the submittal of qualification samples. This report shall include an analysis of all parts when subjected to the most severe natural environmental stresses of this specification. Failure rate data source shall be MIL-HDBK-217, where applicable. Other failure rate data sources, including contractor in-house data, shall be substantiated in the reliability prediction report. The ratios of rated strength under the specified environmental condition to the applied stresses shall be not less than that required to give a predicted MTBF (Mean Time Between Failures) of at least 10,000 hours.

3.7 Life. The charger shall have a minimum life, under all combinations of conditions specified herein, of at least 2000 hours.

3.8 Marking. Markings shall be in accordance with MIL-STD-130.

3.8.1 Nameplate. All chargers shall contain a standard size type A, B, or H metal nameplate in accordance with MIL-P-15024, securely attached to the unit. The nameplate shall contain the following information: Charger, Battery, Nickel-Cadmium, Aircraft

Input: _____ volts DC _____ amperes DC

or

Input: _____ volts AC _____ amperes AC

_____ Hz _____ phase

Output: _____ volts DC max _____ amperes DC max

Military Part No.
National Stock No.
Manufacturer's Part No.
Contract or Order No.
Charger S/N.
Data Mfg.
Manufacturer's name or trademark
Inspector's stamp
US

3.8.2 Part number on nameplate. The manufacturer's part number indicated on the nameplate shall be the number of the contractor's assembly drawing which includes subassembly ordering data.

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3.8.3 Restriction on size of trademark. The manufacturer's name or trademark shall be in letters not larger than other letters appearing on the nameplate.

3.9 Workmanship. Chargers shall be manufactured and assembled in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows.

a. Qualification inspections (see 4.4).

b. Qualification verification inspections (see 4.5).

c. Quality conformance inspections (see 4.6).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.3.1 Inspection system requirements. The inspection system requirements shall be in accordance with MIL-I-45208.

4.4 Qualification inspection. Qualification inspection will be performed by the Government (see 6.3) on samples of chargers furnished by the supplier. Qualification tests shall consist of the examination, inspections and tests specified in Table IV.

4.4.1 Qualification samples. The qualification test sample shall consist of at least three units of each part number for which qualification is sought. The sample units shall be produced, to the maximum extent practicable, with equipment and procedures normally used in production, and shall be identified as specified (see 3.8, 3.8.1 and 3.8.2). Any additional information required by the letter of authorization shall be prepared (see 6.3). Together with the test samples, the manufacturer shall include two reproducible copies of the outline and assembly drawing, schematic, parts list and three printed copies of simple instructions with illustrations and diagrams, if necessary, covering the installation of the charger. Upon completion of qualification testing and prior to

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qualification approval, the manufacturer shall again prepare two reproducible copies of outline and detail assembly drawings, schematics and parts list. Any changes in the reproducible copies from those included with the qualification samples shall be indicated in detail.

4.4.2 Materials certification. Together with the test samples, the manufacturer shall prepare a materials inspection report consisting of certification, supported by verifying data, that the materials and finishes specified in 3.3, used in fabricating the chargers are in accordance with the applicable referenced specifications or specified requirements.

4.5 Verification of qualification. Each charger of the same part number delivered as a qualified item under this specification shall be numbered sequentially in essentially the order that it is submitted for Government acceptance regardless of the contract under which delivered or by whom purchased. To meet the delivery requirement for the first 50 chargers, 51 chargers shall successfully complete the required quality conformance inspections; 50 of which shall be shipped as directed by the applicable contract(s); and one charger, selected at random by the Government Inspector, shall be forwarded to the Qualifying activity (see 6.3). To meet the delivery for each succeeding 150 chargers, 151 chargers shall successfully complete the required quality conformance inspections, 150 of which shall be shipped as directed by the applicable contract(s) and one charger selected at random and forwarded as above. The span of serial numbers and the contract numbers of the production units from which each verification sample was selected shall be recorded and this information included with the sample. The sample shall be shipped no later than the final unit in the production group from which selected. The units forwarded to the qualifying activity will be inspected and tested for conformance to this specification and the applicable detail specification. Tests shall be conducted as shown in Table IV for any one of the qualification samples at the discretion of the qualifying activity. All qualification verification samples become Government property when accepted by the Government Inspector for shipment to the qualifying activity.

4.5.1 Rejection and retest of qualification and qualification verification samples. Samples which have been rejected or returned to the manufacturer for any reason during qualification or qualification verification tests, may be reworked or have parts replaced to correct defects. Before resubmitting the samples, full particulars concerning the rejection and the corrective action taken by the manufacturer must be submitted in writing by the manufacturer to the qualifying activity. Tests shall not be resumed until such a report is received. If the manufacturer does not correct and return components rejected during qualification verification testing within five (5) weeks after he receives them to the qualifying activity or does not submit a program acceptable to the qualifying activity for the correction of deficiencies noted within two (2) weeks the component will be disqualified.

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4.5.2 Disassembly and inspection. At the conclusion of tests, each sample will be disassembled as necessary to inspect for evidence of wear, defects, and overheating. The presence of such evidence shall be cause for rejection.

TABLE IV. Schedule of qualification tests.

Test Title	Charger Sample			Requirement Paragraph	Test Paragraph
	1	2	3		
Examination of Product	X	X	X	3.3, 3.4, 3.7, 3.8, & 3.9	4.7.2.1
Operational Verification	X	X	X	3.5.1	
Efficiency	X			3.5.1.8	4.7.2.2
Overvoltage Surge		X		3.5.1.1	4.7.2.3
High Voltage Spikes		X		3.5.1.1	4.7.2.4.1
Turn-On/Turn-Off Transients		X		3.5.1.3	4.7.2.4.2
Short Circuit/Open Circuit		X	X	3.5.1.7 & 3.5.1.9	4.7.2.4.3
Duty Cycling	X			3.5.3 & 3.7	4.7.2.5
Capacity		X		3.5.1.4.6	4.7.2.7
Operating Position	X			3.5.4	4.7.2.8
Electromagnetic Interference			X	3.5.5	4.7.2.9
Electromagnetic Susceptibility			X	3.5.5	4.7.2.10.1
Acceleration	X			3.5.5	4.7.2.10.2
Dust	X			3.5.5	4.7.2.11
Salt Fog	X			3.5.5	4.7.2.12
Temperature-Altitude		X		3.5.5	4.7.2.13
Mechanical Shock		X		3.5.5	4.7.2.14
Fungus		X		3.5.5	4.7.2.15
Humidity			X	3.5.5	4.7.2.16
Sine Vibration	X			3.5.5	4.7.2.17
Random Vibration			X	3.5.5	4.7.2.18.1
Explosive Atmosphere			X	3.5.5	4.7.2.18.2
Burn-In	X	X	X	3.5.6	4.7.2.19
Disassembly and Inspection	X	X	X	3.5.5	4.7.2.20 4.7.2.21

4.6 Quality conformance inspections. Quality conformance inspections shall consist of inspection of the product for delivery. Each charger offered for delivery shall be subjected to the inspection specified in Table V. Data from these inspections shall be recorded and available for examination by authorized Government representatives.

4.6.1 Noncompliance. If a sample fails to pass quality conformance inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted,

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and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken quality conformance inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Final acceptance and shipment shall be withheld until the quality conformance inspection has shown that the corrective action was successful. In the event of failure after reinspection, further rework and resubmission shall not be allowed.

TABLE V. Quality conformance inspection.

Test Title	Requirement Paragraph	Test Paragraph
Burn-In	3.5.6	4.7.2.20
Examination of Product	3.3, 3.4, 3.7, 3.8 & 3.9	4.7.2.1
Operation Verification	3.5.1	4.7.2.2
Dielectric Strength 1/	3.5.2	4.7.2.6

1/ To be conducted prior to final assembly of the charger. Certification of compliance shall be included when the inspection lot is presented for acceptance.

4.7 Methods of examination and test.

4.7.1 Test conditions. Unless otherwise specified, tests shall be made under the following conditions.

4.7.1.1 Ambient air. Tests shall be performed at an ambient temperature of 23° ± 5°C, at ambient atmospheric pressure and relative humidity.

4.7.1.2 Tolerance and accuracy. The tolerance of test conditions, other than room ambient, and the accuracy of test apparatus shall be as specified in MIL-E-81910.

4.7.1.3 Ambient temperature measurement. When tests are required to be conducted at reduced or elevated temperatures, the ambient temperature shall be determined by measurements of the air at several points surrounding, but not above the unit, and averaging the results.

4.7.1.4 Airflow. For all high temperature-altitude tests below 6.1 kilometers, the air flow over the unit within the test chamber shall be adjusted to approximate natural convection conditions and shall not exceed 30.5 meters/minute while maintaining the required ambient

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temperature within specified tolerance. For tests above 6.1 kilometers altitude, the air flow may exceed 30.5 meters/minute but shall be adjusted to the minimum velocity consistent with maintaining the required ambient temperature within specified tolerance.

4.7.1.5 Battery performance. The batteries used during the evaluation shall be qualified batteries capable of meeting the performance requirements specified herein when charged in accordance with constant-current charge method described in 4.7.1.6.

4.7.1.6 Battery reconditioning. Reconditioning of the battery to balance cells and restore capacity shall be conducted when specified herein. Reconditioning shall consist of subjecting the battery to a 1 C-rate discharge and applying a shorting jumper across each cell's terminals when its potential is 0.5 volt. Following a 24-hour stand, the shorting jumpers shall be removed and the battery shall be charged at the 1 C constant-current rate until the terminal potential reaches 1.55 $\pm$ 0.01 volts per cell, and then at the C/5 rate for 3 more hours. The electrolyte level shall be adjusted within 2-4 hours after charging. Perform 1 C-rate capacity discharge to cutoff voltage (0.95 volt/cell).

4.7.1.7 Position. The units shall be mounted base down.

4.7.1.8 Input voltage. Nominal input voltage shall be applied as indicated on the applicable specification sheet.

4.7.2 Methods of inspection.

4.7.2.1 Visual and mechanical inspection. The charger shall be inspected to determine compliance with the requirements of this specification and the applicable specification sheet with respect to materials, design and construction, parts, workmanship, dimensions, weight, and markings, for which there are no applicable tests.

4.7.2.2 Operation verification. The charger shall be connected to a nickel-cadmium battery of known capacity that has been in a shorted condition for 12-24 hours. With input power applied to the charger, it shall charge the battery to automatic shutoff. Upon completion of charge, the battery shall be disconnected form the charger and allowed to stand open circuit for 1 hour. The battery shall then be discharged at the 1 C rate, and the capacity of the battery determined. The capacity must not be less than 95 percent of its predetermined value (see 4.7.1.6).

4.7.2.2.1 The type II charger shall operate with line-to-neutral input voltages between 108 and 118 volts. During charge operation, the frequency shall be varied from 360 to 440 Hz. Also, the charger shall be operated for 1.0 minute with 115 V 1Ø 400 Hz power applied. The type I charger shall operate with input voltages of 23.0 volts to 29.0 volts. For input terminal voltages of 26.0 - 29.0 volts dc or 112 - 118 volts 3Ø 400 Hz, charging current shall be sufficient to meet all charge requirements specified herein, and shall meet the charger input and output current requirements specified on the applicable specification sheet.

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4.7.2.2.2 Temperature sensor. The charger temperature sensing circuit shall be evaluated for battery protection by simulating the resistance of the temperature sensor specified on the applicable specification sheet. With the charger operating at its rated load and nominal input power, the external resistance applied to the charger's temperature sensing input terminals shall be gradually increased from the temperature sensor's nominal resistance at 55°C to its nominal resistance at 70°C, and then to its nominal resistance at 50°C. The charger shall comply with 3.5.1.4.1 and 3.5.1.7. With the charger operating at its rated load and nominal input power, the external resistance applied to the charger's temperature sensing input terminals shall be gradually decreased from the temperature sensor's nominal resistance at -25°C to its nominal resistance at -40°C, and then to its nominal resistance at -25°C. The charger shall comply with 3.5.1.4.1 and 3.5.1.7. With the charger operating at its rated load and nominal input conditions, open circuit and short circuit the temperature sensor input terminals of the charger. The charger shall comply with 3.5.1.4.1.1 and 3.5.1.7.

4.7.2.2.3 Thermal switch. The charger thermal switch sensing circuit shall be evaluated for battery overtemperature protection by simulating a temperature sensor failure and closing of the battery thermal switch. The external resistance applied to the charger's temperature sensing input terminals shall be the temperature sensor's nominal resistance at 25°C. With the charger operating at its rated load and nominal input conditions, apply a ground signal to the input terminal of the thermal switch for 5 minutes, then open circuit the input terminal of the thermal switch for 5 minutes. The charger shall comply to 3.5.1.4.2 and 3.5.1.7.

4.7.2.2.4 Cell imbalance. The charger cell imbalance circuit shall be evaluated during all modes of charger operation by simulating a cell imbalance condition within the battery. With nominal input power applied, and in all modes of charger operation, apply a variable external reference voltage to the center tap voltage sensing input. Vary the voltage to simulate a cell imbalance condition in each battery half. Record the voltage of the simulated cell imbalance condition for each battery half, which causes the charger to turn off. The charger shall comply with 3.5.1.4.3 and 3.5.1.7.

4.7.2.2.5 Charge inhibit. The charge inhibit circuit shall be evaluated, with the charger operating at its rated load and nominal input conditions, by applying external inhibit signals to the charger, as defined on the applicable specification sheet. The charger shall comply with 3.5.1.5, 3.5.1.6, and 3.5.1.7.

4.7.2.2.6 Battery relay control. The battery relay control circuit shall be evaluated, to verify that the type I chargers provide proper control of the aircraft's battery relay.

a. With the charger operating at its rated load and nominal input conditions, verify that the charger's battery relay control contacts, as defined on the applicable specification sheets, are open.

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b. Reduce input voltage to 20.0 V for 1-2 seconds, then increase input voltage to 26.3 V. Verify that the relay control contacts close, and charging stops, when the input voltage is momentarily reduced to 21.0 ± 1.0 V. The relay contacts shall remain closed for 4.5 ± 2.5 seconds, regardless of input voltage, and reopen when the input voltage is increased to 26.0 ± 0.3 V.

c. Reduce input voltage to 20.0 V for 5 minutes, and verify that the battery relay control contacts remain closed. Increase input voltage to 28.0 V and verify that the relay contacts open at 26.0 ± 0.3 V. Increase input voltage to 30.0 V and verify that battery relay control contacts remain open.

d. Verify that the battery relay control contacts close upon application of a charge inhibit signal, and open upon removal of the charge inhibit signal. The charger shall comply with 3.5.1.6 and 3.5.1.7.

4.7.2.3 Efficiency. Discharge battery specified on applicable specification sheet to 0.95 volt/cell. With nominal input power applied to the charger and at conditions of 4.7.1.1, charge battery to automatic shutoff. Measure input and output power with average reading meters 15 minutes after start of charge. Efficiency shall conform to 3.5.1.8.

4.7.2.4 Input voltage transients.

4.7.2.4.1 Overvoltage surge. Each of the following transients shall be applied to the charger while the charger is operating and connected to a completely discharged battery, and also, while the charger is operating and connected to a fully charged battery. The unit shall conform to 3.5.1.1.

a. Type I

50 volts for 12.5 milliseconds
46 volts for 25 milliseconds
40 volts for 47.5 milliseconds
30 volts for 77.5 milliseconds
29 volts for 1.0 minute
22 volts for 1.0 minute

b. Type II

180 volts for 10 milliseconds
160 volts for 34 milliseconds
140 volts for 60 milliseconds
124 volts for 80 milliseconds
100 volts for 1.0 minute

The charger shall not exceed the input and output current requirements specified on the applicable specification sheet and shall not be damaged during this test.

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4.7.2.4.2 High voltage spikes. High voltage spikes shall be applied to the input terminals by a spike generator having an effective internal impedance of 50 ± 5 ohms. The open circuit spike voltage waveform shall conform with MIL-STD-461 except that the spike voltage shall not exceed ± 200 volts. Spikes shall be injected in accordance with MIL-STD-462, Method CS06. One hundred spikes of each polarity shall be applied at a rate of 10 pulses per second or less for each of the following operating modes:

a. Open circuit.

b. Charge.

4.7.2.4.3 Turn-on/turn-off transients. With the charger connected to the battery, operating at its rated load and nominal input conditions, turn off the input power. With the charger connected to the battery, turn on nominal input power. With the charger connected to the battery, operating at its rated load and nominal input conditions, disconnect battery from charger. With input power applied to charger, connect battery to charger. The charger shall meet the requirements of 3.5.1.3.

4.7.2.5 Short circuit/open circuit test. With nominal input power applied, open circuit the charger output terminals for 1 hour. During the 1 hour period, the input voltage shall be varied between 23.0 and 29.0 VDC for type I chargers, and between 108 and 118 VAC for type II chargers. Following the open circuit test, with nominal input power applied, the charger output terminals shall be shorted for 1 hour. During this 1 hour period, the input voltage shall be varied between 23.0 and 29.0 VDC for type I chargers, and between 108 and 118 VAC for type II chargers. There shall be no damage to the charger and the charger shall pass the test of 4.7.2.2 immediately following this test.

4.7.2.6 Dielectric strength. Dielectric strength tests shall be conducted on components and subassemblies, as necessary, before final assembly to determine the parts' ability to meet all specified dielectric requirements.

4.7.2.6.1 Circuits. Circuits which operate at voltage no greater than 50 volts shall be tested at 500 volts (rms), 60 Hz for 1 minute or 600 volts (rms), 60 Hz for 1 second. Circuits operating in excess of 50 volts dc shall be tested at a minimum of 10 times the nominal rated potential for 1 second.

4.7.2.6.2 Capacitors. Capacitors shall be tested for 1 minute at a voltage equal to twice the voltage and at the pulse repetition rate expected to be encountered in the intended application (see 6.4.1).

4.7.2.7 Duty cycle test. The charger duty cycle test shall be conducted at ambient conditions of 4.7.1.1 in conjunction with the battery specified on the applicable specification sheet. The charger shall be instrumented with thermocouples mounted on temperature generating

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and temperature sensitive components within the charger. The battery-charger system shall be subjected to a duty cycle test consisting of automatic charge/discharge cycles. Charger and battery temperatures shall be monitored throughout the test. Temperatures of critical components within the charger shall be used to verify the life of the charger, as specified in 3.7. The battery-charger system shall be tested as follows:

a. Conditioning. Conduct battery reconditioning procedure (see 4.7.1.6).

b. Duty cycles.

(1) Charge battery at 1 C constant-current rate until the terminal potential reaches 1.55 ± 0.01 volts per cell, then at the C/5 rate for 3 more hours. The electrolyte level shall be adjusted within 2-4 hours after charging.

(2) Rest on open circuit for 2 hours.

(3) Discharge through a fixed resistance of

$$R = \frac{0.0315n}{C}$$ Where:

R = ohms

n = number of cells

C = rated capacity in ampere-hours.

Discharge for 20 seconds.
Rest on open circuit for 260 seconds.
Discharge for 20 seconds.
Total time 300 seconds.

Voltage readings shall be taken at end of each discharge.

(4) Within 3 ± 1 seconds after end of second discharge, apply nominal input power to charger and charge battery for 1 hour and 55 minutes. Repeat (2) above, repeat (3) above.

(5) Repeat (4) continuously for 54 cycles total.

c. Cycle 55 - With nominal input power applied to charger, charge battery for 2 hours. Discharge battery at C rate to 0.95 volt/cell. The battery capacity must not be less than 85 percent of its value determined during conditioning (see 4.7.1.6).

d. Cycle 56 - Repeat c. above.

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e. With nominal input power applied to charger, charge battery for 2 hours. The electrolyte level shall be adjusted within 2 hours after charging. Water consumption shall not exceed 0.5 cc/cell per ampere-hour of capacity, e.g., 10 cc/cell for 20 ah battery, for the 56-cycle test. Battery reconditioning shall not be permitted during cycles 1 through 56.

f. The test shall be terminated when any one of the following conditions occur:

(1) The 20-second voltage is below 0.6315 volt/cell during any duty cycle discharge.

(2) Discharge requirements of cycles 55 and 56 not met.

(3) The successful completion of 56 cycles.

4.7.2.8 Capacity. Prior to conducting these tests, the battery shall be subjected to reconditioning cycles as necessary to balance the cells (see 4.7.1.6). The battery and charger shall be tested to demonstrate compliance with the 25°C requirements of Tables I and II. The charge/discharge cycle shall be conducted as follows, starting with a battery discharged to 0.95 volt/cell.

a. Charge for the specified 80 percent charge time (see 3.5.1.4.5).

b. Open circuit stand for a period of 1 to 24 hours.

c. Discharge at the rate specified in Table II to the specified end voltage. Discharge time shall not be less than that specified in Table II for 80 percent state-of-charge.

d. Allow battery to stand open circuit until the battery test temperature has stabilized.

e. Charge for the specified 100 percent charge time (see 3.5.1.4.5).

f. Open circuit stand for a period of 1 to 24 hours.

g. Discharge at the rate specified in Table II to the specified end voltage. Discharge time shall not be less than that specified in Table II for 100 percent state-of-charge.

4.7.2.8.1 49°C capacity. The charger, not operating and with no circuits energized, and appropriate battery shall be soaked at 49°C for a minimum of 16 hours. The charger and battery shall then be tested

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in accordance with 4.7.2.8 to determine compliance with the 49°C requirement of Tables I and II.

4.7.2.8.2 -18°C capacity. The charger, not operating and with no circuits energized, and appropriate battery shall be soaked at -18°C for a minimum of 16 hours. The charger and battery shall then be tested in accordance with 4.7.2.8 to determine compliance with the -18°C requirement of Tables I and II.

4.7.2.8.3 -29°C capacity. The charger, not operating and with no circuits energized, and appropriate battery shall be soaked at -29°C for a minimum of 16 hours. The charger and battery shall then be tested in accordance with 4.7.2.8 to determine compliance with the -29°C requirement of Tables I and II.

4.7.2.9 Operating position. With the charger operating in its main mode and nominal input conditions, place charger on each of its six surfaces for 20 minutes each. Confirm that operation of the charger in any position does not affect power input, heating, or other operating characteristics.

4.7.2.10 Electromagnetic compatibility.

4.7.2.10.1 Electromagnetic interference. Conducted and radiated electromagnetic interference shall be measured in accordance with MIL-E-81910. The charger shall be tested during all modes of operation, and shall conform to the emission requirements of CE03 and RE02, as contained in MIL-STD-461, except that the limits of MIL-STD-461 are raised 20 db across the test frequency range.

4.7.2.10.2 Electromagnetic susceptibility. The charger shall be subjected to electromagnetic susceptibility tests in accordance with MIL-E-81910. The charger shall be tested during all modes of operation.

4.7.2.11 Acceleration. The charger shall be tested in accordance with MIL-E-81910. During Procedure II, the charger shall operate at its rated load with nominal input power applied. The output current shall be monitored during the test and shall be free of abnormal interruptions and transients. At the conclusion of the test, the charger shall be examined for compliance with 3.5.5, and shall pass the test of 4.7.2.2.

4.7.2.12 Dust. The charger shall be tested in accordance with MIL-E-81910, except that no input power shall be applied during the test. At the conclusion of the test, the charger shall pass the test of 4.7.2.2.

4.7.2.13 Salt fog. The charger shall be tested in accordance with MIL-E-81910. The charger shall stand at room ambient conditions for a period of 24 hours, and shall then pass the test of 4.7.2.2. The test of 4.7.2.2 shall be repeated after completion of an additional 48-hour stand at room temperature. The charger shall be examined for compliance with 3.5.5.

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4.7.2.14 Temperature-altitude. The charger shall be tested in accordance with MIL-E-81910 for self-cooled equipment, except that the maximum and minimum test temperatures shall be 71°C and -55°C, respectively. The maximum test altitude shall be 15.24 kilometers. The charger shall be electrically evaluated at 3.048-kilometer intervals by connecting it to a discharged battery located outside of the test chamber, and charging the battery to automatic shutoff. The charger shall comply with 3.5.1.4. At the conclusion of the test the charger shall pass the test of 4.7.2.2.

4.7.2.15 Mechanical shock. The charger shall be tested in accordance with MIL-E-81910, except that the equipment crash safety test, Procedure III of MIL-STD-810 shall not be performed. At the conclusion of the test, the charger shall be examined for compliance with 3.5.5, and shall pass the test of 4.7.2.2.

4.7.2.16 Fungus. The charger shall be tested in accordance with MIL-E-81910. At the conclusion of the test, the charger shall be inspected for fungus growth and shall pass the test of 4.7.2.2.

4.7.2.17 Humidity. The charger shall be tested in accordance with MIL-E-81910. The charger shall charge the appropriate size battery from a discharged state to automatic shutoff while the charger is at 30°C during each cycle of the humidity test. The charger shall comply with 3.5.1.4. At the conclusion of the test, the charger shall be examined for compliance with 3.5.5, and shall pass the test of 4.7.2.2.

4.7.2.18 Vibration.

4.7.2.18.1 Sine vibration. The charger shall be tested in accordance with MIL-E-81910 for non-engine-mounted components, except that the sine vibration test displacements, acceleration levels and crossover frequencies shall be as follows:

5-10 Hz	5.08 mm Double Amplitude (DA)
10-23 Hz	1 g
23-74 Hz	0.9144 mm DA
74-2000 Hz	10 g

During each axis of vibration, the charger shall charge the appropriate size battery from a discharge state to automatic shutoff. The charger shall comply with 3.5.1.4, and the charger output waveform shall be free of abnormal interruptions and transients. At the conclusion of the test, the charger shall be examined for compliance with 3.5.5, and shall pass the test of 4.7.2.2.

4.7.2.18.2 Random vibration. The charger shall be tested in accordance with MIL-E-81910 for non-engine-mounted components, except that the charger shall only be subjected to the functional level of random vibration. During each axis of vibration the charger shall charge the appropriate size battery from a discharged state. The charger shall comply with 3.5.1.4 and the charger output waveform shall be free

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of abnormal interruptions and transients. At the conclusion of the test, the charger shall be examined for compliance with 3.5.5 and shall pass the test of 4.7.2.2.

4.7.2.19 Explosive atmosphere. The charger shall be subjected to the explosive atmosphere test of MIL-STD-810, Method 511, Procedure I. The maximum chamber temperature shall not exceed 60°C and the maximum test altitude shall be 15.24 kilometers. During the test, the charger shall be operated to actuate all making and breaking of electrical contacts. All thermal switches which are not actuated under the specified test conditions shall be tested separately at temperatures which will cause them to actuate. Ignition of the explosive mixture shall be cause for rejection.

4.7.2.20 Burn-in. The charger shall be subjected to the burn-in test in accordance with MIL-E-81910. During the burn-in test the charger shall be subjected to a cycling regime described as follows:

a. Nominal input voltage shall be applied continuously.

b. The charger shall be operated in the main charge mode for 1 hour.

c. The charger shall be operated in the topping and trickle charge modes for 3 hours.

d. This 4-hour cycling regime shall be performed twice during each burn-in cycle.

4.7.2.21 Disassembly and inspection. At the conclusion of the qualification inspection, the chargers shall be disassembled and inspected for evidence of excessive deterioration and defects (see 3.5.5).

5. PACKAGING

5.1 Application. The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified by the procuring activity, each charger shall be packaged in accordance with MIL-STD-794. Preservation shall be in accordance with MIL-P-116, Method II, without preservation compound. One charger shall be packed per container.

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5.2.2 Level C. Each charger shall be individually packaged in a manner which affords adequate protection against deterioration and physical damage during shipment in accordance with the manufacturer's commercial practice when such meets the requirements of this level.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.3.1 Level A. Unless otherwise specified by the procuring activity, chargers shall be packed in accordance with MIL-STD-794. Shipping containers shall be exterior export type conforming to Tables II or III of MIL-STD-794.

5.3.2 Level B. Unless otherwise specified by the procuring activity, chargers shall be packed in accordance with MIL-STD-794. Shipping containers shall be exterior domestic type conforming to Tables II or III of MIL-STD-794.

5.3.3 Level C. Shipping containers shall meet the requirements of MIL-STD-794 for level C packing.

5.4 Marking. In addition to any special marking required by the contract or order, each package and shipping container shall be durably and legibly marked in accordance with MIL-STD-129.

5.5 Acceptance stamp. Each package and shipping container shall be durably and legibly marked in accordance with the requirements of MIL-STD-129. In addition, two opposite sides of each shipping container or exterior packing shall be marked with the Government acceptance stamp.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. The battery chargers covered by this specification are intended for use in aircraft electrical systems for the purpose of charging nickel-cadmium aircraft batteries.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Title, number and date of the applicable specification sheet (see 3.1).

c. Level of preservation, packaging and packing required for shipment (see Section 5).

d. Number of samples to be submitted for inspection and type inspection to be performed thereon.

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e. Marking and shipping (see Section 5).

f. Place of final inspection.

g. Installation instructions, handbooks, or manuals when required.

h. That the contractor shall include the cost of units required for verification of qualification (see 4.5) in the amount of his bid.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command (Attention: Commanding Officer, Naval Air Engineering Center (ESSD), (Code 93), Lakehurst, NJ 08733), however information pertaining to qualification of products may be obtained from the qualifying activity or the Naval Weapons Support Center (3052), Crane, Indiana 47522.

6.3.1 Engineering changes to a qualified item. Once a manufacturer's part number has been placed on the Qualified Products List, the manufacturer is responsible for any changes to the charger which affect its physical or performance characteristics. No changes should be made to a qualified charger without the specific written approval of the qualifying agency. The manufacturer should allow sufficient lead time for the qualifying agency to repeat the qualification tests if necessary.

6.4 Miscellaneous notes.

6.4.1 Dielectric strength. The dielectric strength test for capacitors described in Section 4 is not a test of ultimate dielectric strength and is not intended to be a design criterion for capacitors. The purpose of this test is to assure the satisfactory performance of individual capacitors under operating conditions. Capacitors used shall be suitable for the purpose and shall conform to applicable Government specifications.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 82 MAR PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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